Comparing Two Populations

Chapter 10, with a slice of Chapter 11

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Objectives

- Choose the right two-sample procedure from the **design**: independent samples, paired samples, or two proportions.
- Compare two means from independent samples with Welch's t-test (the default) and know when the pooled test applies.
- Explain why pairing makes a test more powerful, and analyse matched pairs as a one-sample test on the differences.
- Compare two proportions with a test (pooled) and an interval (unpooled).
- Use "does the interval for the difference contain zero?" to read a two-sample result.

- Size an A/B test, and read its result as a manager should.
- Tell an experiment from an observational comparison, and say what a significant difference does and does not let you conclude.
- Explain why testing many pairs of groups inflates false positives, and read a one-way ANOVA table.

Week 9 built one engine: an estimate, its standard error, and a reference distribution. This week every question compares **two** groups: federal vs. private salaries, end-aisle vs. middle-aisle displays, version A of a web page vs. version B. The engine does not change. Every test in the chapter has the same shape:

$$\frac{(\text{difference in the estimates}) - (\text{difference claimed in } H_0)}{\text{standard error of the difference}}.$$

What changes is the standard error, and **the design of the study decides which one**. Were the two groups sampled independently? Was the same store or customer measured twice? Are we comparing averages or percentages? Get the design right and the formula follows. Get it wrong and a real effect can vanish, as the Rockstar example shows.

The note ends with a slice of Chapter 11: what goes wrong when you compare many groups two at a time, and the one test (ANOVA) built to avoid it.

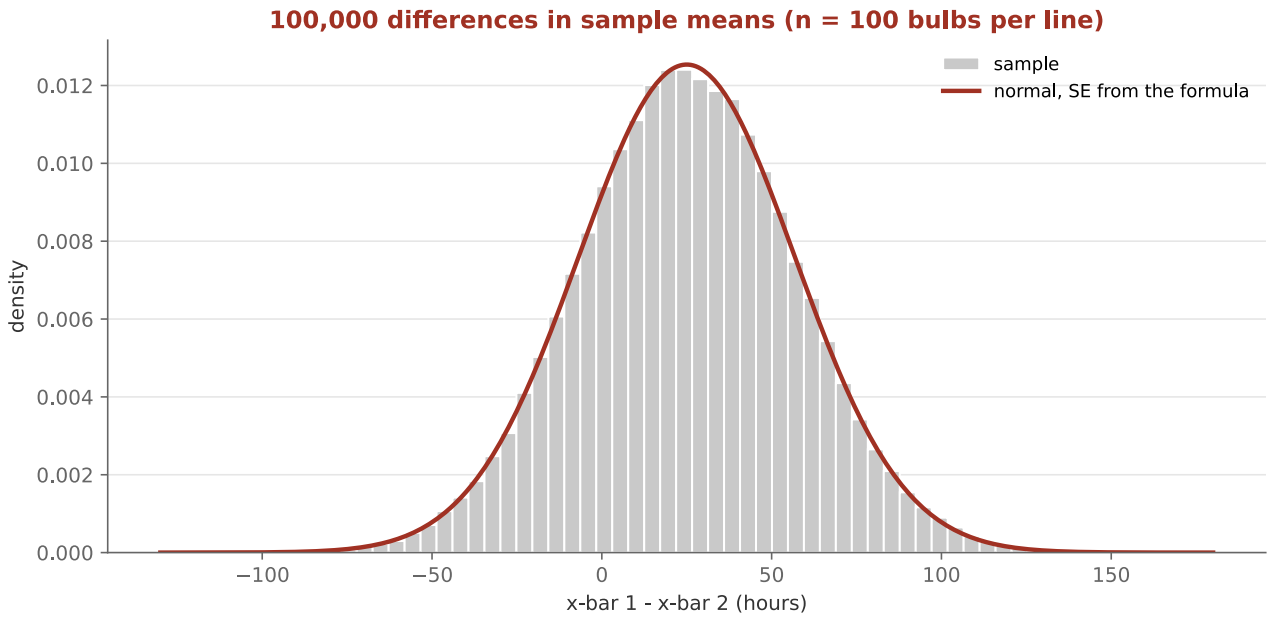
Two Independent Samples

The sampling distribution of a difference

For two **independent samples**, where what is drawn from one population has no bearing on what is drawn from the other, the difference in sample means $\bar{x}_1 - \bar{x}_2$ has

$$\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2, \qquad \sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

The second quantity is the **standard error of the difference**. Variances add, even though the means subtract: two uncertain numbers make their difference *more* uncertain, never less.



Two production lines of light bulbs, with true means 8,025 and 8,000 hours and $\sigma = 225$ on each line. The simulated differences have standard deviation 31.88; the formula gives $\sqrt{225^2/100 + 225^2/100} = 31.82$. Notice how often the difference comes out *negative* even though line 1 really is better. A 25-hour edge is barely one standard error, which is why the old version of this note found a Type II error with exactly these settings.

Define population 1 and population 2 before writing any hypothesis. The choice is arbitrary, but every sign that follows depends on it. With federal workers as population 1, "federal pays more" is $\mu_1 - \mu_2 > 0$; reverse the labels and it becomes $\mu_1 - \mu_2 < 0$. Most sign errors in this chapter start here.

Sigmas unknown: Welch's test is the default

In practice the population standard deviations are unknown, and the sample standard deviations take their place. That brings in the t-distribution, and a choice.

Welch's two-sample t-test (unequal variances) uses

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)_{H_0}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

with degrees of freedom from the **Welch–Satterthwaite** formula (the textbook's Formula 10.11):

$$\mathrm{df} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}.$$

It makes **no assumption** that the two populations have the same variance. The df always falls between the smaller of $n_1 - 1$ and $n_2 - 1$ and the pooled test's $n_1 + n_2 - 2$; it is closer to the low end when one sample is much noisier or much smaller than the other.

The **pooled t-test** assumes $\sigma_1 = \sigma_2$ and combines the two sample variances into one estimate of the common variance, the **pooled variance**

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

Its standard error of the difference and test statistic are

$$\mathrm{SE}_{\bar{x}_1 - \bar{x}_2} = \sqrt{s_p^2}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, \qquad t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)_{H_0}}{\sqrt{s_p^2}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

with $n_1 + n_2 - 2$ degrees of freedom. This is Welch's standard error with s_p^2 in place of both s_1^2 and s_2^2: $\sqrt{s_p^2/n_1 + s_p^2/n_2} = \sqrt{s_p^2}\sqrt{1/n_1 + 1/n_2}$.

Use Welch unless you have a reason not to. When the variances really are equal, Welch gives nearly the same answer as the pooled test. When they are not, the pooled test can be badly wrong, especially with unequal sample sizes. Your textbook gives the same advice ("when in doubt … proceed with the unequal variances test"), and it is the default in most modern software. Excel offers both, side by side, in **Data → Data Analysis**: *t-Test: Two-Sample Assuming Unequal Variances* (Welch) and *… Assuming Equal Variances* (pooled).

Application: air conditioning or open windows?

A consumer group drives a Nissan Maxima on one tank of highway fuel, 10 drivers with the air conditioning on and windows up, 11 with the AC off and windows down. Is there a difference in fuel economy (mpg)? Population 1 is AC on.

$$H_0 : \mu_1 - \mu_2 = 0 \qquad H_1 : \mu_1 - \mu_2 \neq 0 \qquad \alpha = 0.05$$

	value
n1	10
s1	1.9311
n2	11
s2	4.4998
method	Welch (unequal variances)
x-bar 1	27.47
x-bar 2	24.6364
difference x-bar1 - x-bar2	2.8336
H0 difference	0
alternative	two-sided
standard error of the difference	1.4878
df	13.8313
test statistic t	1.9045
critical value	+/- 2.1472
p-value	0.0779
alpha	0.05
reject H0?	False
Excel tool	t-Test: Two-Sample Assuming Unequal Variances

The AC-on cars averaged 2.83 mpg more, $t = 1.90$ with 13.83 degrees of freedom, and the two-tailed p-value is 0.0779. Above 0.05: **the data are not sufficient evidence that fuel economy differs** between the two ways of driving. The 95% interval for the difference, −0.36 to 6.03 mpg, contains zero, which says the same thing and shows how wide the uncertainty is.

Notice the two standard deviations: 1.93 and 4.50. Windows-down mileage is more than twice as variable, so assuming equal variances would be hard to defend. (The pooled test gives p = 0.0815 here, the same conclusion.)

$$\text{df} = \frac{(0.3729 + 1.8408)^2}{\frac{0.3729^2}{9} + \frac{1.8408^2}{10}} = \frac{4.9004}{0.0155 + 0.3388} = 13.83$$

About those degrees of freedom. For the Maxima data, $s_1^2/n_1 = 0.3729$ and $s_2^2/n_2 = 1.8408$, and the Welch–Satterthwaite formula above gives a fraction, 13.83. That is closer to the windows-down sample's own $n_2 - 1 = 10$ than to the pooled test's 19, because the noisier sample contributes most of the standard error. Software uses the fraction. Excel's Data Analysis tool rounds it to the nearest whole number (14). Your

textbook rounds *down* by hand (13), to be conservative. The three p-values differ only in the third decimal place, so any of them is acceptable; on exams, read the one the output gives you.

Application: a pooled test, in an experiment

In the textbook's rat study, rats from one population were assigned to an enriched environment or a barren one, and brain weights were compared after three months: enriched $\bar{x} = 2.6$, $s = 0.6$, $n = 20$; barren $\bar{x} = 2.1$, $s = 0.8$, $n = 25$. Does enrichment increase brain weight? The book assumes equal variances, so we use the pooled test, and show Welch beside it.

$$H_0 : \mu_1 - \mu_2 \le 0 \qquad H_1 : \mu_1 - \mu_2 > 0 \qquad \alpha = 0.05$$

	pooled (book)	Welch
x-bar 1	2.6	2.6
x-bar 2	2.1	2.1
difference x-bar1 - x-bar2	0.5	0.5
H0 difference	0	0
alternative	greater	greater
pooled variance	0.5163	
standard error of the difference	0.2156	0.2088
df	43	42.8537
test statistic t	2.3196	2.3946
critical value	1.6811	1.6812
p-value	0.0126	0.0105
alpha	0.05	0.05
reject H0?	True	True

$$s_p^2 = \frac{(20-1)(0.6)^2 + (25-1)(0.8)^2}{20+25-2} = \frac{6.84 + 15.36}{43} = 0.5163$$

$$\text{SE} = \sqrt{0.5163}\sqrt{\frac{1}{20} + \frac{1}{25}} = 0.7185 \times 0.3000 = 0.2156$$

$$t = \frac{(2.6 - 2.1) - 0}{0.2156} = 2.3196, \qquad \text{df} = 20 + 25 - 2 = 43$$

Pooled: $t = 2.32$, $p = 0.0126$. Welch: $t = 2.39$, $p = 0.0105$. Either way we reject H_0: **the data are evidence that the enriched environment produces heavier brains on average.** And because the rats were *randomly assigned* to environments, this is a rare case where "produces" is justified. We return to that word below.

Does the interval for the difference contain zero?

A confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm (\text{critical value}) \times (\text{standard error of the difference}).$$

If it **excludes 0**, a two-tailed test at $\alpha = 1 - \text{confidence}$ rejects "no difference"; if it **includes 0**, the test does not. This is week 9's interval–test link, applied to a difference.

When both population standard deviations are known (rare, but it is where the textbook starts), the critical value comes from z. A sample of 35 federal jobs averages \$66,700 and 32 comparable private-sector jobs average \$60,400, with $\sigma_1 = \$12,000$ and $\sigma_2 = \$11,000$ taken as known.

		value	interval
difference x-bar1 - x-bar2	6300		
H0 difference	0		
standard error of the difference	2809.8996		
test statistic z	2.2421		
critical value	+/- 1.9600		1.96
p-value	0.025		
alpha	0.05		
reject H0?	True		
Excel tool	z-Test: Two Sample for Means		
difference			6300
standard error			2809.8996
df			z
margin of error			5507.302
lower			792.698
upper			11807.302
contains 0?			False

$z = 2.24$, $p = 0.0250$, and the 95% interval for the difference is \$793 to \$11,807. It excludes zero, so the test rejects. The interval says more than the test: federal salaries are higher on average, by an amount we can pin down only to somewhere between about \$800 and \$11,800. That is a wide range for a policy argument, and it is the interval, not the p-value, that shows it.

Testing a difference other than zero

Sometimes a difference only matters if it exceeds a threshold. A Ruby Tuesday manager will change the music only if slow-music evenings raise the average bill by **more than \$3.00** per person. Slow music: $\bar{x} = \$45.90$, $s = \$5.60$, $n = 23$; fast music: $\bar{x} = \$39.25$, $s = \$5.20$, $n = 19$.

$$H_0 : \mu_1 - \mu_2 \leq 3 \qquad H_1 : \mu_1 - \mu_2 > 3$$

	value
method	pooled (equal variances)
x-bar 1	45.9
x-bar 2	39.25
difference x-bar1 - x-bar2	6.65

	value
H0 difference	3
alternative	greater
pooled variance	29.416
standard error of the difference	1.6814
df	40
test statistic t	2.1708
critical value	1.6839
p-value	0.018
alpha	0.05
reject H0?	True
Excel tool	t-Test: Two-Sample Assuming Equal Variances

The numerator subtracts the \$3.00 claimed in H_0: $t = \frac{(45.90-39.25)-3.00}{\text{SE}} = 2.17$, $p = 0.0180$ (Welch: 0.0174). Reject: **the data are evidence that slow music raises the average bill by more than \$3.00.** In Excel, type 3 into *Hypothesized Mean Difference*.

Paired Samples

Two samples are **dependent**, or **paired**, when each observation in one is matched to a specific observation in the other: the same store in two weeks, the same customer before and after, the same golfer with two balls. A **matched-pairs test** works with the differences $d_i = x_{1i} - x_{2i}$ and runs a **one-sample t-test** on them:

$$t = \frac{\bar{d} - \mu_{d,H_0}}{s_d/\sqrt{n}}, \qquad \text{df} = n - 1,$$

where n is the number of **pairs**. Hypotheses are about μ_d, the population mean difference, never about $\bar{d}$.

Application: where to put the energy drinks

Josh, a PepsiCo account manager, wants to know whether Rockstar sells more on an end-of-aisle display than on a middle-aisle shelf. In each of nine stores he records one week of sales with each placement. Population 1 is end-aisle, so $d = \text{end} - \text{middle}$.

$$H_0 : \mu_d \le 0 \qquad H_1 : \mu_d > 0 \qquad \alpha = 0.05$$

store	1	2	3	4	5	6	7	8	9
end aisle	64	54	126	97	37	74	117	90	81
middle aisle	72	41	100	62	40	60	122	62	78
difference d	−8	13	26	35	−3	14	−5	28	3

	value
correlation of the pairs r	0.8418
pairs n	9

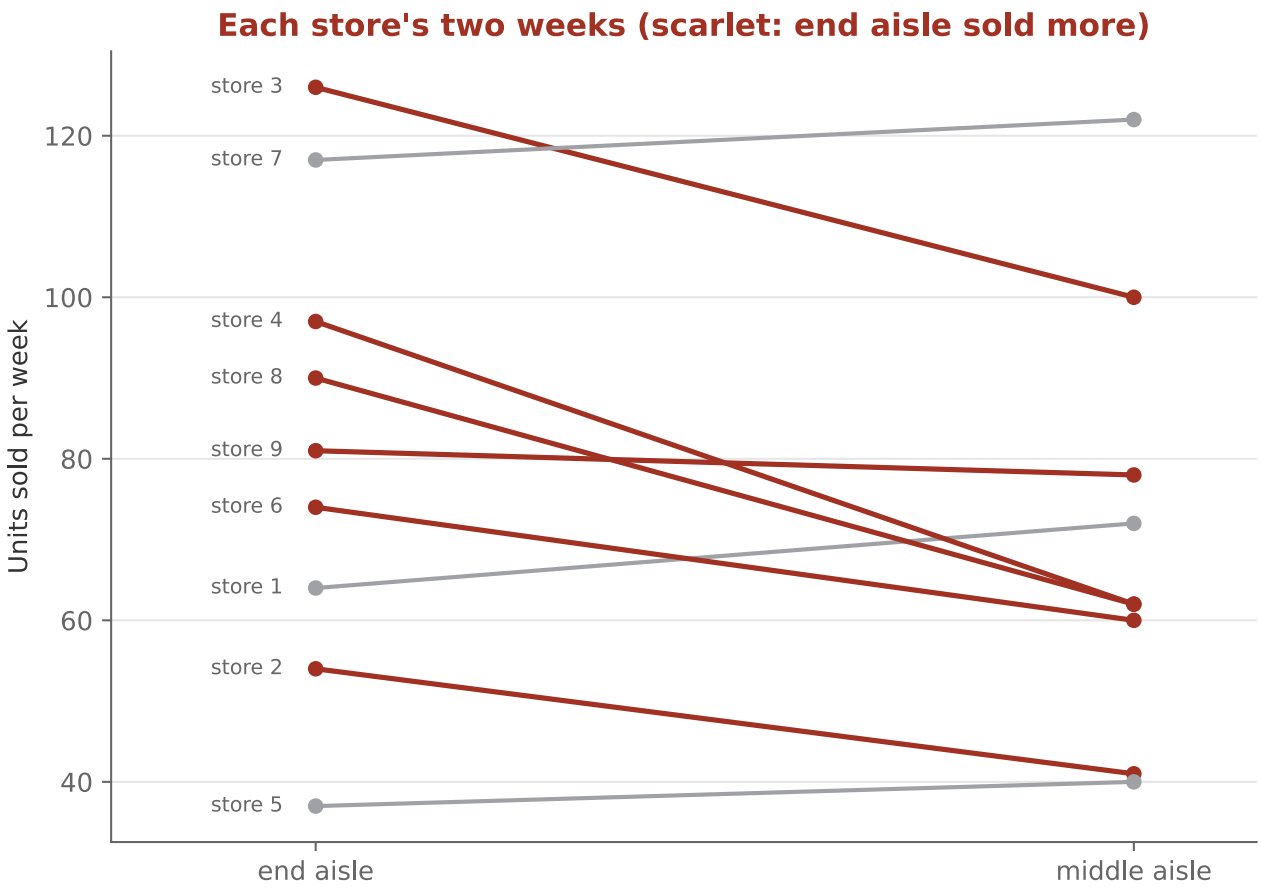
	value
mean difference d-bar	11.4444
s_d	15.7251
standard error s_d/sqrt(n)	5.2417
df	8
test statistic t	2.1834
critical value	1.8595
p-value	0.0303
alpha	0.05
reject H0?	True
90% interval for mu_d: lower	1.6973
90% interval for mu_d: upper	21.1916
Excel tool	t-Test: Paired Two Sample for Means
Excel p-value	=T.DIST.RT(2.1834, 8)

> Across the nine stores, end-aisle displays sold 11.44 more units a week on average ($s_d = 15.73$), $t = 2.18$ with 8 df, $p = 0.0303$. Reject H_0: **the data are evidence that end-aisle placement sells more on average.** The 90% interval for the mean gain is 1.7 to 21.2 units a week.
>
> Why 90%? Week 9: a one-tailed test at $\alpha = 0.05$ matches a two-sided **90%** interval, not a 95% one. The 95% interval here, −0.6 to 23.5, just reaches below zero. That is not a contradiction; it says the evidence is real but modest, and that a two-sided question ("does placement matter in either direction?") would not quite clear the bar with nine stores.

Why pairing matters: the same numbers, analysed wrongly

Now pretend the 18 numbers came from 18 *different* stores and run Welch's independent-samples test on them.

	paired (correct design)	independent (ignores the pairing)
difference	11.4444	11.4444
standard error	5.2417	13.0613
df	8.0000	15.8921
t	2.1834	0.8762
p-value	0.0303	0.1970



	value
s1^2	830.94
s2^2	704.44
r	0.84
−2 r s1 s2	−1288.11
Var(d) = sum	247.28
Var(d) directly	247.28
Var(d) if the pairing is ignored (r = 0)	1535.39

The same difference, 11.44 units, gives $p = 0.030$ analysed as pairs and $p = 0.197$ analysed as independent samples. One analysis finds the effect; the other misses it entirely.

The chart shows why. Store 3 sells about 100+ units whatever the placement; store 5 sells about 40. That store-to-store spread is huge, and an independent-samples test counts all of it as noise. But it is the *same* spread in both weeks: the two columns are strongly correlated ($r = 0.84$). Taking each store's difference cancels it. In numbers,

$$\mathrm{Var}(d) = s_1^2 + s_2^2 - 2 r s_1 s_2,$$

which is 247 here, against 1535 if the pairing is ignored. **Pairing removes variation you already know about, so the variation you care about can show through.** It is the most powerful idea in this chapter, and it returns at the end of the note as *blocking*.

The analysis must match the design. Paired data analysed as independent throws away power, as above. Independent data analysed as paired is simply wrong (which observation would you subtract from which?). Ask: was each observation in one group matched to a *specific* observation in the other? If yes, pair.

A design flaw worth spotting. In the Rockstar study, every store ran the end-aisle week *first*. If demand was rising (warmer weather, a holiday, a promotion), the first week would look better for reasons that have nothing to do with placement. A better design flips a coin in each store to decide which placement goes first. Statistics can measure a difference; only the design can tell you what caused it.

A golf-ball maker claims its new ball adds **more than 15 yards** to a golfer's drive. Eight golfers each hit the new ball and their own ball:

Golfer	1	2	3	4	5	6	7	8
New ball	201	228	256	233	248	255	239	220
Own ball	204	213	231	241	223	226	212	186

(a) State the hypotheses. **(b)** Test at $\alpha = 0.05$. **(c)** Give a 90% interval for the mean gain. **(d)** What must be assumed with only 8 pairs?

i Solution

	value
correlation of the pairs r	0.6303
pairs n	8
mean difference d-bar	18
s_d	15.5012
standard error s_d/sqrt(n)	5.4805
df	7
test statistic t	0.5474
critical value	1.8946
p-value	0.3006
alpha	0.05
reject H0?	False
90% interval for mu_d: lower	7.6168
90% interval for mu_d: upper	28.3832
Excel tool	t-Test: Paired Two Sample for Means
Excel p-value	=T.DIST.RT(0.5474, 7)

(a) $d = \text{new} - \text{own}$; $H_0 : \mu_d \le 15$, $H_1 : \mu_d > 15$ (the claim needs evidence). **(b)** $\bar{d} = 18.00$ yards, $t = 0.55$, $p = 0.3006$. Do not reject H0: the average gain is above 15 yards in this sample, but 8 golfers are not enough evidence that the true average gain exceeds 15. **(c)** 7.6 to 28.4 yards. **(d)** The differences should be roughly normal, since 8 pairs are far too few for the Central Limit Theorem.

Comparing Two Proportions

For x_1 successes in n_1 and x_2 in n_2, with $\hat{p}_1 = x_1/n_1$ and $\hat{p}_2 = x_2/n_2$:

Test of $H_0 : p_1 = p_2$. H_0 says there is one common proportion, so estimate it by pooling both samples, $\bar{p} = \frac{x_1+x_2}{n_1+n_2}$, and use

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}.$$

Confidence interval for $p_1 - p_2$. The interval claims nothing, so each sample keeps its own proportion:

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}.$$

This is the two-sample version of week 9's rule: **the test uses what H_0 claims, the interval uses what the data show.** In week 9 that meant p_0 for the test and $\hat{p}$ for the interval; here it means the pooled $\bar{p}$ for the test and the separate $\hat{p}$'s for the interval.

Application: homeownership in two regions

105 of 150 sampled southeastern homes and 80 of 125 northeastern homes are owner-occupied. Population 1 is the Southeast. Do the rates differ?

	test (pooled)	interval (unpooled)
p-hat 1	0.7	0.7
p-hat 2	0.64	0.64
difference	0.06	0.06
H0 difference	0.0	
pooled p-bar	0.6727	
standard error	0.0568	
test statistic z	1.0559	
critical value	+/- 1.9600	
p-value	0.291	
alpha	0.05	
reject H0?	False	
Excel p-value	=2 * (1 - NORM.S.DIST(1.0559, TRUE))	
standard error (unpooled)		0.0569
critical value z		1.96
margin of error		0.1116
lower		−0.0516
upper		0.1716
contains 0?		True

> The sample rates are 70% and 64%. Pooled, $\bar{p} = 0.673$, $z = 1.06$, $p = 0.291$: **no evidence that homeownership rates differ** between the regions. The 95% interval, −0.052 to 0.172, contains zero and runs from the Northeast being about 5 points higher to the Southeast being about 17 points higher. With samples this size, even a 6-point gap is within the noise.

Application: do good students have fewer accidents?

Liberty Mutual gives young drivers a good-student discount. In one state, 167 of 1,207 good students had accidents, against 799 of 4,963 other young drivers.

$$H_0 : p_1 - p_2 \geq 0 \qquad H_1 : p_1 - p_2 < 0$$

	value
p-hat 1	0.1384
p-hat 2	0.161
difference	−0.0226
H0 difference	0.0
pooled p-bar	0.1566

	value
standard error	0.0117
test statistic z	−1.9406
critical value	−1.6449
p-value	0.0262
alpha	0.05
reject H0?	True
Excel p-value	=NORM.S.DIST(-1.9406, TRUE)

Accident rates of 13.8% and 16.1%, $z = -1.94$, one-tailed $p = 0.0262$. Reject: **good students have a lower accident rate.** The difference is about 2.3 percentage points (95% interval 0.1 to 4.5 points lower).

That justifies the discount as **pricing**: an insurer prices risk, and a lower accident rate is lower risk, whatever the reason. It does **not** show that good grades make anyone a safer driver. Good students may drive less, drive newer cars, or have parents who supervise more. The next section is about exactly this difference.

Experiments and Observational Comparisons

In an **experiment**, the investigator assigns the units to the groups, ideally at random. In an **observational comparison**, the groups come as they are: the investigator only records which group each unit belongs to.

Example	Who chose the group?	What a significant difference shows
Rats, enriched vs. barren cages	the researchers, at random	enrichment **causes** heavier brains
Rockstar, end vs. middle aisle	Josh, but always end-aisle first	placement *probably* matters; order is a rival explanation
Federal vs. private salaries	the workers themselves	the averages differ; not *why* (education, location, job mix)
Good students vs. others	the students themselves	the accident rates differ; not that grades cause safe driving
A/B test of two web pages	random assignment of visitors	the page version **causes** the difference in conversions

A two-sample test tells you whether the groups differ, not why. Random assignment is what licenses "causes": it makes the groups alike, on average, in every respect except the one you changed, including the ones you never measured. Without it, every difference comes with rival explanations, and a tiny p-value does nothing to rule them out. This is the most important judgment call you will make with this chapter's tools, and it returns in the regression weeks as "correlation is not causation."

A/B Testing

An **A/B test** is a two-proportion test run as an experiment: visitors are randomly shown version A or version B of a page, an email, or an offer, and the conversion rates are compared. It is the most common use of this chapter in business today, and the random assignment is what makes its conclusions causal.

Each group in an experiment is called an **arm**, a term borrowed from clinical trials. A drug trial randomly assigns patients to a **treatment arm**, which gets the new drug, and a **control arm**, which gets a placebo or the current standard of care. An A/B test has the same structure:

- the **control arm** (A) sees the current page;
- the **treatment arm** (B) sees the new design.

Sample sizes are stated **per arm**: the number of visitors who must see *each* page. A test that needs 3,841 per arm needs 3,841 on page A **and** 3,841 on page B, 7,682 in all.

Thinking of an A/B test as a small clinical trial is a good habit. A trial asks the questions this chapter has been building toward: Who decides which patient gets the drug? (A coin, not the doctor, so the arms are alike in everything else.) How many patients does each arm need to detect a benefit worth having? (A power calculation, done in advance.) When is the trial over? (When the planned number is reached, not when the results first look good.) Ask the same three questions of any A/B test you are shown.

How many visitors?

The question to answer *before* launch is week 9's power question. Suppose the current page converts 10% of visitors and a lift to 12% would be worth the redesign.

Sample size per arm for an A/B test (a two-sided, two-proportion test):

$$n = \frac{\left(z_{\alpha/2}\sqrt{2\bar{p}(1-\bar{p})} + z_{\beta}\sqrt{p_1(1-p_1)+p_2(1-p_2)}\right)^2}{(p_2-p_1)^2}, \qquad \bar{p} = \frac{p_1+p_2}{2},$$

rounded **up**. This is a *planning* calculation, done before any data are collected, so none of its inputs come from the test's samples:

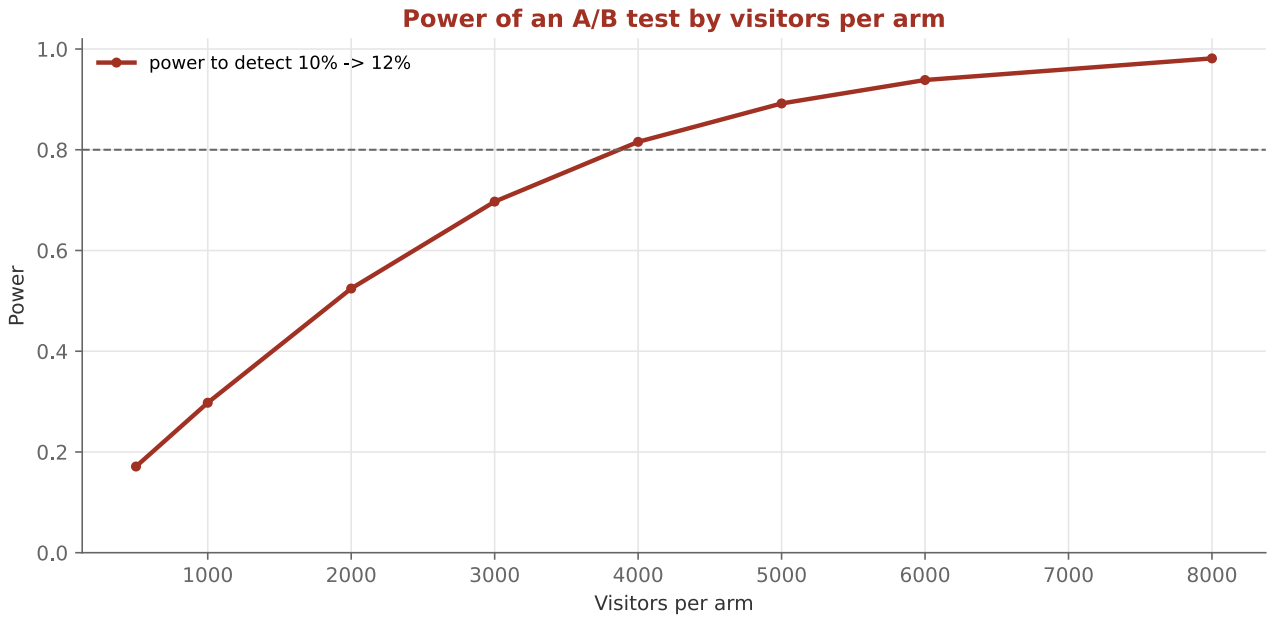
- p_1 is the **baseline** rate, known from the page's history;
- p_2 is the baseline plus the **smallest lift worth detecting**, chosen by the business, so $\delta = p_2 - p_1$ is a choice, not a measurement;
- $z_{\alpha/2}$ sets the false positive rate α, and z_{β} sets the power $1-\beta$ (the chance of detecting a lift of size δ if it is really there).

The first term in the numerator is the noise in $\hat{p}_1 - \hat{p}_2$ if H_0 is true; the second is the noise if the lift is real.

$$n = \frac{\left(1.9600\sqrt{2(0.11)(0.89)} + 0.8416\sqrt{0.10(0.90)+0.12(0.88)}\right)^2}{(0.12-0.10)^2} = \frac{(0.86727+0.37222)^2}{0.0004} = \frac{1.53634}{0.0004} = 3,840.85$$

	10% vs 12%
baseline p1	0.1000
target p2	0.1200
lift (points)	2.0000
alpha	0.0500
power	0.8000
raw n per arm	3840.8475
n per arm (round UP)	3841.0000

	10% vs 12%
total visitors	7682.0000



Detecting a 2-point lift with 80% power takes **3,841 visitors per arm**, 7,682 in all. With 1,000 per arm the test would catch a real lift only 30% of the time. A 1-point lift (10% to 11%) needs 14,751 per arm. Small lifts are expensive to detect, which is why the size of the lift worth detecting has to be decided by the business before the test starts.

Why 80% power, and what it implies

Eighty percent power is the conventional choice: if the lift is really there, the test has an 80% chance of rejecting H_0 and a 20% chance (β) of missing it. Paired with $\alpha = 0.05$, it treats a false positive as about four times as costly as a false negative, which suits the usual situation: a false positive commits money to a redesign that does not work, while a false negative only means the idea gets tested again later. It is a convention, not a law. Raise it to 90% when missing a real gain is expensive, and remember that every step up in power costs visitors.

Four quantities are tied together: α, power, the lift δ, and n. Fix any three and the formula determines the fourth. Above we fixed α, power, and δ, and solved for n. Run it the other way and a sample size you can afford *implies* the smallest lift you can reliably detect, the **minimum detectable effect**:

visitors per arm	smallest detectable lift (points), 80% power	smallest detectable lift (points), 90% power
1000	4.07	4.77
2000	2.82	3.29
3841	2.00	2.33
8000	1.37	1.59
15000	0.99	1.15

With 1,000 visitors per arm, a test at $\alpha = 0.05$ with 80% power can reliably detect only a lift of about 4.1 points (10% to 14.1%). A smaller real lift may well be there; this test would usually miss it. So α and power alone do *not* pin down δ: it is α, power, **and** n together that imply the detectable lift. The business decides which one to solve for.

The sample size depends on the question you ask. Each chapter added a requirement, and each requirement costs visitors. All rows use a 10% baseline, a 2-point difference, and 95% confidence ($\alpha = 0.05$, two-sided).

Chapter	Formula	Question it answers	n (per arm in Ch 10)
8	$z^2 p(1-p)/E^2, p = 0.10$	Estimate **one** rate to within ± 2 points	865
8	same, $p = 0.12$	Same, planning with the higher rate	1,015
9	$\left((z_{\alpha/2} + z_\beta)\sigma/\delta\right)^2, \sigma = \sqrt{0.1(0.9)}$	Test **one** sample against a **known** 10%, 80% power	1,766
10	A/B formula with $z_\beta = 0$	Compare **two estimated** rates, 50% power	1,881
10	A/B formula above	Compare two estimated rates, **80% power**	**3,841**

Two steps account for the jump. **Estimating both rates** roughly doubles the variance of the difference: in Chapters 8 and 9 the 10% baseline was treated as a known number, but in an A/B test the control group's rate is just as noisy as the variant's. **Requiring power** adds the z_β term: a study sized only so that a 2-point difference would be "significant" detects a real 2-point lift about half the time. None of the earlier formulas is wrong; each answers a smaller question.

Three rules for reading an A/B test.

1. **Fix the sample size in advance, and don't stop when p first dips below 0.05.** Checking every day and stopping at the first "significant" reading is a way of running many tests and keeping the best one; the false positive rate climbs far above 5%.
2. **Report the interval for the lift, not just the p-value.** "B converts 0.3 to 2.1 points better" supports a decision; "$p = 0.03$" does not.
3. **One primary metric, decided in advance.** Testing ten metrics and reporting the one that "won" is the multiple comparisons problem at the end of this note.

An online retailer randomly sends 2,400 customers email A and 2,400 email B. A produces 168 purchases, B 216. **(a)** Test at $\alpha = 0.05$ whether the purchase rates differ. **(b)** Give a 95% interval for B's lift over A. **(c)** Can the retailer say email B *caused* the extra purchases? Why?

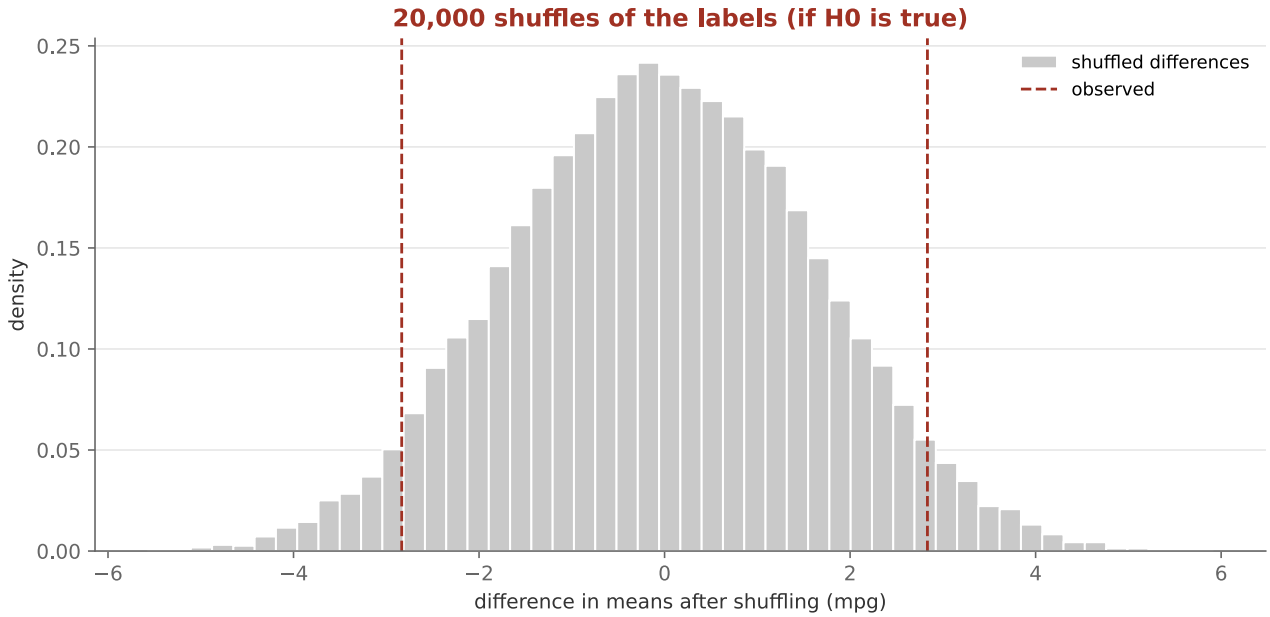
Solution

	test	interval
p-hat 1	0.09	0.09
p-hat 2	0.07	0.07
difference	0.02	0.02
H0 difference	0.0	
pooled p-bar	0.08	
standard error	0.0078	
test statistic z	2.5538	
critical value	+/- 1.9600	
p-value	0.0107	
alpha	0.05	
reject H0?	True	
Excel p-value	=2 * (1 - NORM.S.DIST(2.5538, TRUE))	
standard error (unpooled)		0.0078
critical value z		1.96
margin of error		0.0153
lower		0.0047
upper		0.0353
contains 0?		False

Population 1 is email B. **(a)** Rates 9.0% vs. 7.0%, $z = 2.55$, $p = 0.0107$. Reject H0: the rates differ. **(b)** 0.5 to 3.5 percentage points. **(c)** Yes, within the limits of the test: customers were *randomly* assigned to the emails, so the groups were alike in everything except the email.

A Test With No Formula: Shuffling the Labels

If the group labels made no difference, the 21 Maxima mileages could have been split into “AC on” and “windows down” in any way at all. So shuffle the labels thousands of times, recompute the difference in means each time, and see how often chance alone produces a gap as big as the real one.



	Maxima, two-sided
observed difference	2.8336
shuffles	20000.0000
permutation p-value	0.0813

8.1% of the shuffles produce a difference at least as large as the observed 2.83 mpg in either direction. That share *is* a p-value, and it agrees closely with Welch's 0.0779. No formula, no normality assumption, no degrees of freedom: just the definition of a p-value, done by brute force. When a formula and a permutation test agree, you can trust both; when they disagree (small, skewed samples), trust the permutation test.

Which Test Is It?

situation	statistic	standard error	df	Excel
Two means, independent, sigmas known (rare)	z	sqrt(s1^2/n1 + s2^2/n2) with sigmas	-	z-Test: Two Sample for Means
Two means, independent, sigmas unknown (DEFAULT)	t (Welch)	sqrt(s1^2/n1 + s2^2/n2)	Welch formula	t-Test: Two-Sample Assuming Unequal Variances
Two means, independent, variances assumed equal	t (pooled)	sqrt(sp^2 (1/n1 + 1/n2))	n1 + n2 - 2	t-Test: Two-Sample Assuming Equal Variances
Two means, paired (same units twice)	t on d = x1 - x2	s_d / sqrt(n)	n - 1 (pairs)	t-Test: Paired Two Sample for Means
Two proportions, test of p1 = p2	z	sqrt(p-bar (1 - p-bar)(1/n1 + 1/n2)), pooled	-	NORM.S.DIST on the z

situation	statistic	standard error	df	Excel
Two proportions, confidence interval	z	sqrt(p1(1-p1)/n1 + p2(1-p2)/n2), unpooled	-	NORM.S.INV for the critical value
Three or more means	F (one-way ANOVA)	MS within	k - 1 and N - k	Anova: Single Factor

Name the procedure and write the hypotheses.

(a) A casino compares the average weekly play of 60 loyalty members before and after a new rewards tier.

(b) A bank compares the share of under-40 and over-40 customers who bank mainly online, from separate random samples.

(c) An HR analyst compares average commute times of 45 Henderson and 38 Summerlin employees, and the Summerlin times are far more variable.

(d) A chain tests three store layouts across 30 stores.

ℹ Solution

(a) Paired t (the same members twice): $H_0 : \mu_d = 0$ (or one-sided, if the question is whether play rose). **(b)** Two-proportion z, pooled for the test: $H_0 : p_1 = p_2$. Observational, so a difference would not show that age *causes* online banking. **(c)** Welch two-sample t: $H_0 : \mu_1 = \mu_2$. **(d)** Three groups: one-way ANOVA (below), not three t-tests.

Reading the Textbook: Corrections

1. **"If the populations are not normally distributed, the t-distribution can still be used … if both sample sizes are 30 or more. This is because the shape of the t-distribution approaches the normal."** Large samples help because the Central Limit Theorem acts on $\bar{x}_1 - \bar{x}_2$, not because t becomes z. Strongly skewed data may need more than 30 per group (week 9's income simulation).
2. **Welch degrees of freedom "rounded up … is a Type I error."** Rounding up would raise the Type I error *rate* slightly; it is not itself an error. Software uses the exact fraction, Excel rounds to the nearest whole number, the book rounds down. Any of the three is acceptable.
3. **Rockstar's design.** End-aisle always came first, so a trend in demand is a rival explanation (above).
4. **Causal language on observational data.** The federal-salary and good-student conclusions show that the groups differ, not why.
5. **"Because α = 0.05 is within this range, we can reject"** (Rockstar, Step 8). The table brackets p between 0.025 and 0.05; the exact value, 0.030, settles it. Use exact p-values from Excel.

Many Groups: A Slice of Chapter 11

Chapter 11 extends this week's question to three or more groups: three store layouts, four suppliers, five ad campaigns. We will not compute its sums of squares by hand. Three ideas from it are worth your time.

Idea 1: comparing many pairs inflates false positives

The obvious approach to four groups is to run a t-test on every pair. Four groups make six pairs, each tested at $\alpha = 0.05$. If all four population means are in fact equal, what is the chance that at least one of the six tests cries wolf?

groups k	pairwise tests m	P(at least one false positive)	Bonferroni alpha per test
2	1	0.0500	0.0500
3	3	0.1426	0.0167
4	6	0.2649	0.0083
5	10	0.4013	0.0050
6	15	0.5367	0.0033
8	28	0.7622	0.0018
10	45	0.9006	0.0011

	4 identical groups of 20, simulated 5,000 times
groups	4
per group	20
simulations	5000
pairwise tests each time	6
share with >= 1 'significant' pair	0.2192
share where ANOVA rejects	0.0536
nominal alpha	0.05

Your textbook's arithmetic: with six independent tests, the chance of at least one false positive is $1 - 0.95^6 = 0.265$. With ten groups (45 pairs) it is 90%. The simulation draws four groups from the *same* population, so every "significant" difference is false, and still finds at least one significant pair 21.9% of the time. (The six tests share data, so they are not independent and the rate is a little below the formula's.) One ANOVA test on the same data rejects 5.4% of the time, as it should.

This is week 9's false-positive problem scaled up, and it is everywhere in business: dashboards that flag "significant" changes in dozens of metrics, A/B tests read segment by segment, "we sliced the data twelve ways and found something." **The more comparisons you make, the more false positives you should expect.** The remedies are to decide on the comparison in advance, to use one test built for many groups (ANOVA), or to demand a stricter α for each comparison (the Bonferroni rule divides α by the number of tests, the last column above).

Idea 2: one-way ANOVA compares many means with one test

One-way analysis of variance (ANOVA) tests

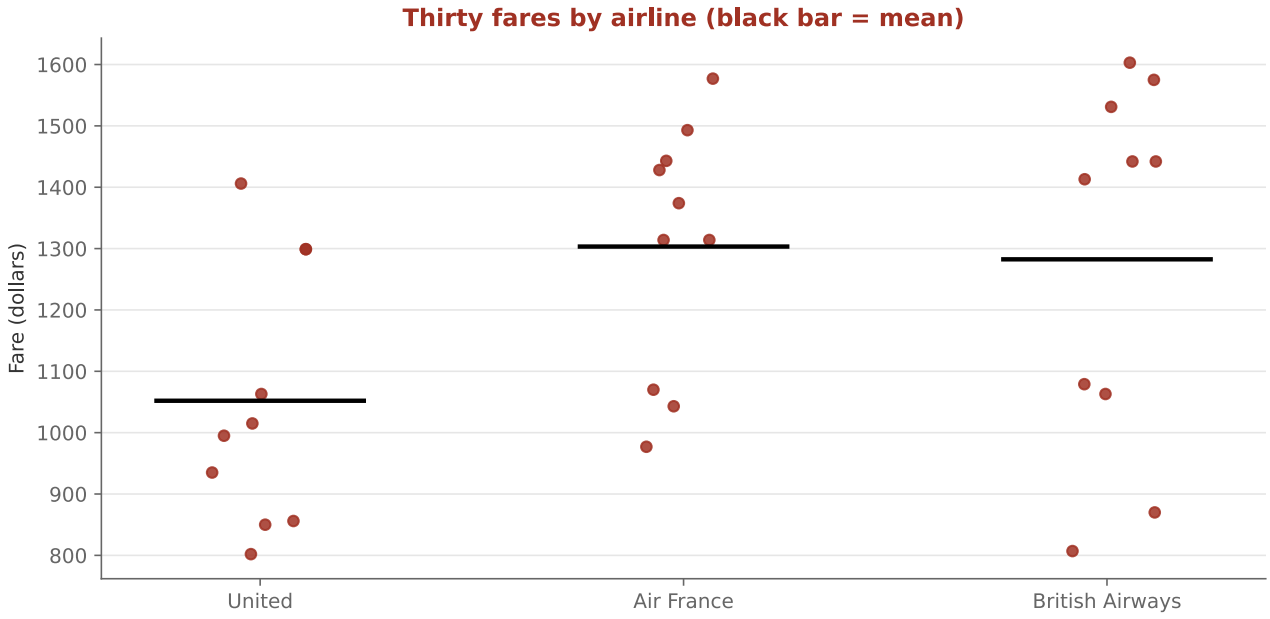
$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \qquad H_1 : \text{not all the means are equal}$$

with a single statistic,

$$F = \frac{\text{variation between the group means (per df)}}{\text{variation within the groups (per df)}} = \frac{MS_{\text{between}}}{MS_{\text{within}}}.$$

If H_0 is true, the group means differ only by noise and F is usually near 1. A large F says the means differ by more than the noise within the groups can explain. The test is always upper-tailed.

The textbook's airfare data record 30 fares between Chicago and London, 10 on each of three airlines (half of each airline's fares in each direction). Do average fares differ by airline?



	count	sum	average	variance
United	10	10520.0	1052.0	45275.8
Air France	10	13033.0	1303.3	42205.3
British Airways	10	12825.0	1282.5	89309.8

	SS	df	MS	F	P-value	F crit
Between Groups	389048.6	2	194524.3	3.3009	0.0522	3.3541
Within Groups	1591118.6	27	58930.3185			
Total	1980167.2	29				

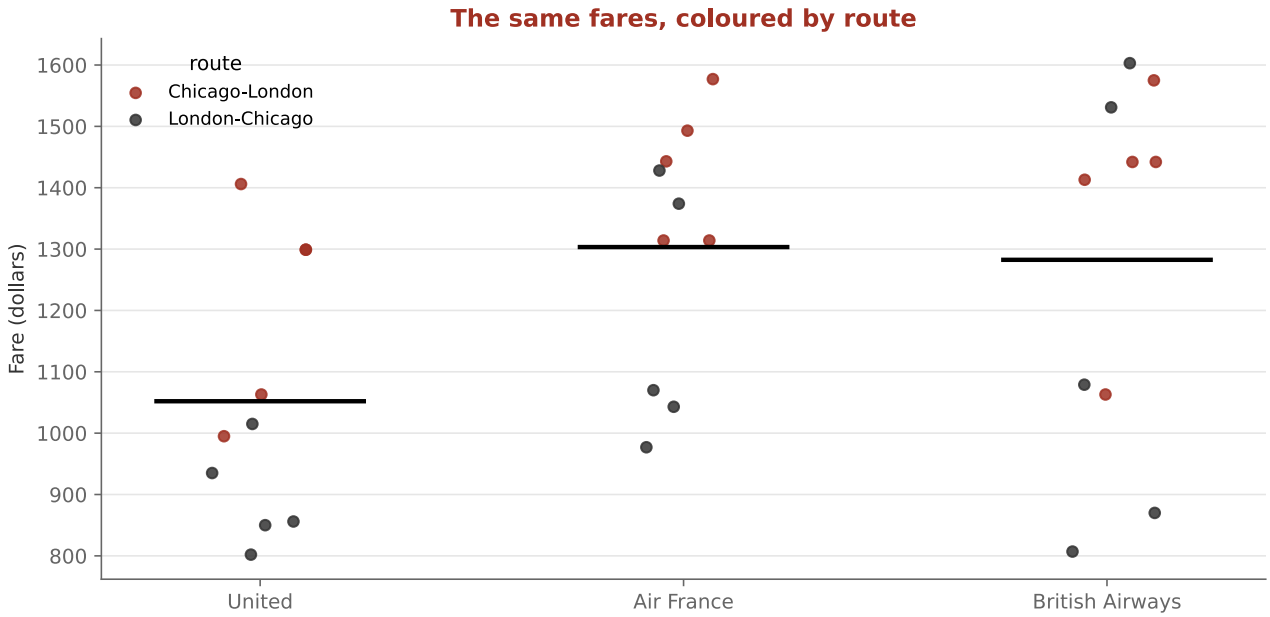
This is the layout of Excel's **Data → Data Analysis → Anova: Single Factor**. Read it in three steps. **Between groups** is how far the three airline means sit from the overall mean; **within groups** is how much fares vary inside each airline; F is their ratio per degree of freedom, here 3.30. The p-value is 0.0522, just above 0.05 (equivalently, F is just below the critical value 3.35). **The data are not quite sufficient evidence that average fares differ by airline.**

Two cautions. A rejection says only that *some* means differ, not which ones; a follow-up procedure (the textbook's Tukey–Kramer test) answers that while holding the overall false positive rate at α. And ANOVA assumes roughly normal data with similar spreads in each group, the same conditions as the pooled t-test.

ANOVA with two groups is the pooled t-test. Run a one-way ANOVA on the same 30 fares grouped by *route* (Chicago–London vs. London–Chicago): $F = 9.6231$. The pooled two-sample t on the same data is 3.1021, and $t^2 = 9.6231$. Same test, same p-value (0.0044). ANOVA is this week's test, generalized to more than two groups.

Idea 3: blocking is pairing for many groups

Look at the strip chart again, now coloured by the direction of travel.



Inside every airline, the Chicago–London fares average well above the London–Chicago fares, by $209 to $321. One-way ANOVA counted that route-to-route spread as noise, exactly as the independent-samples test counted Rockstar's store-to-store spread as noise. Treat route as a **block**, a known source of variation to be removed, and test airline again:

	SS	df	MS	F	P-value	F crit
airline (factor)	3.890486e+05	2	194524.3	4.663	0.0186	3.369
route (block)	5.064801e+05	1	506480.1333	12.1409	0.0018	4.2252
Error	1.084638e+06	26	41716.8641			
Total	1.980167e+06	29				

With route removed from the error term, the airline effect has $F = 4.66$ and $p = 0.0186$: now **the data are evidence that average fares differ by airline.** Nothing about the airlines changed between the two analyses; the noise got smaller. This is the Rockstar lesson again. Pairing compared each store with itself; blocking compares each airline within the same route. Your textbook calls this design a *randomized block ANOVA*.

What we leave for later. Chapter 11's two-way ANOVA also tests whether the *airline* effect depends on the *route* (an **interaction**; here it does not, $p \approx 0.84$). Interactions are easier to understand as terms in a regression model, and that is where we will meet them. The ANOVA table itself also returns: every regression output you read in the coming weeks begins with one, and its F-test asks the same question as here.

Laboratory 10

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account, including the free tier, and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 10 pack — one file. Unzip it and you will have a folder holding 7 files. Drag all 7 into a new Claude conversation, wait for each to show as attached, then paste this:

PROMPT

I have uploaded 7 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.

Then run `10_comparing_two_populations.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

If web access is working on your account, Claude can retrieve the files itself:

PROMPT

Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 10 and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.

Then run `10_comparing_two_populations.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach the page, **nothing is wrong with your work or with the repository.** Claude's web access is a setting, and even when it is on, Claude cannot always retrieve a file by direct link. Use Path 1 instead — it is the ordinary route, not a consolation prize.

Your own junior analyst

PROMPT

A hotel tested a new check-in process. For 12 front-desk agents I have each agent's average check-in time (minutes) in the week before and the week after: before = 6.1, 5.4, 7.2, 4.8, 6.6, 5.9, 7.8, 5.1, 6.4, 5.7, 6.9, 5.5; after = 5.2, 5.0, 6.1, 4.9, 5.5, 5.2, 6.6, 4.6, 5.8, 5.3, 6.0, 5.1. By running code, (1) test whether the new process reduced average check-in time at alpha = 0.05, using the design these data came from, (2) run the WRONG test (treating the weeks as independent samples) and show me how the answer changes and why, and (3) give a 95% interval for the average time saved. Then tell me whether this study shows the new process CAUSED the improvement, and what you would change about the design to make that claim stronger.

Check four things in the reply: the paired test uses 11 degrees of freedom; the explanation of the wrong test mentions the correlation between an agent's two weeks; the interval is for the mean *difference*; and the answer to the last question raises the before/after design itself (a learning curve, a quieter week, or the agents knowing they were watched could all produce the improvement).

Working in Excel

PROMPT

> Build me an Excel workbook that teaches why pairing matters. Sheet 1 has 20 stores. Column B is each store's "size", =NORM.INV(RAND(), 100, 30). Column C is week-1 sales = B + NORM.INV(RAND(), 0, 8), and column D is week-2 sales = B + 5 + NORM.INV(RAND(), 0, 8), so week 2 is truly 5 units better. At the top, put the paired p-value =T.TEST(C2:C21, D2:D21, 2, 1), the unequal-variance p-value =T.TEST(C2:C21, D2:D21, 2, 3), and the correlation =CORREL(C2:C21, D2:D21) side by side. (Use these formulas rather than the Data Analysis tool, whose output does not recalculate.) Give me the file to download.

Press F9 ten times and record both p-values each time. Then change the 30 in column B to 5 and repeat. In one paragraph: when does pairing help a lot, when does it hardly matter, and what does the correlation have to do with it?